

TLA+ Study Group — Week 01

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Brief Description of the Topics

Lecture 1

What is TLA+, for what can it be used and how it's helpful.

- A tool for specification, finding and correcting design errors.
- Well suited for parallel and distributed programming.
- TLA+ sits at the design level/above the code level.
- It abstracts less-critical/lower level parts of systems.
- The properties TLA+ can check are conditions on individual executions (i.e 99% of executions are correct is not a property testable by TLA+)
- Execution of a system is represented as a sequence of discrete steps
- TLA+ represents systems as state machines.

What stood out

... not solutions but recognizing **important** problems that had been obscured by irrelevant details

- The process of simplicifaction by removing irrelevant details is called abstraction.
- *TLA+ has added significant value, either finding subtle bugs we would not have found by other means, or giving us enough confidence to make aggressive optimizations without sacrificing correctness*
- *We witnessed first hand the brain washing done by years of C programming – amen*
- Parallel and distributed systems being represented by sequential processes is kinda obvious (Parallel Turing Machines are exactly as powerful as Turing Machines) yet profound.

Open Questions